1 The pairing gap (M21)

Let $\{(q_i, a_i)\}_{i=1}^{n}$ be question–answer pairs and $S_{ij} = -\log p_\theta(a_j \mid q_i)$ the summed loss of answer a_j after question q_i under the model θ, with D the number of answer tokens. For a cyclic rotation π of the answers,

$$\Delta_\pi \;=\; \frac{1}{D}\sum_{i=1}^{n}\big(S_{i\pi(i)} - S_{ii}\big), \qquad \bar{\Delta} \;=\; \mathbb{E}_\pi\,\Delta_\pi, \tag{1}$$

in nats per answer token, estimated with four seeded rotations and no training. Writing $S_{ij} = r_i + c_j + e_{ij}$, the row and column effects cancel for every π, and averaging over the $n-1$ rotations gives $\bar{\Delta} = -\frac{n}{n-1}\frac{1}{D}\sum_i e_{ii}$: M21 is the interaction between questions and their own answers. It is invariant to any change that depends only on the question (a broken prompt format, a suppression of the topic) or only on the answer (vocabulary, length, a penalty on forbidden outputs). In description-length terms, $D\bar{\Delta}$ is the code length the model already saves on the pairings before training, the head start of the elicitation curve of Donoway et al. [4] at $n = 0$. By Jensen's inequality the n-way identification information of any read-out is bounded by $\bar{\Delta}$ per item, so a gap at its floor means no read-out can pick the answers above chance. We read M21 against two references: a model that cannot hold the pairings sets the floor (TinyStories-1B on WMDP-bio reads 0.22), and on free text, where absolute values do not transfer across tasks, we report $r = \bar{\Delta}_B/\bar{\Delta}_{\text{own}}$, the fraction of the gap the model shows on its own answers to the same questions; $r \geq 0.5$ reads elicit-side. M21 is a sufficient sign of held information, not a necessary one: information that no conditional likelihood reflects reads 0 and can still be elicited.

2 Unlearning: the capability stays in the weights

We read M21 on held-out WMDP-bio facts [6] for Zephyr-7B and six unlearned versions, and for the Deep Ignorance suite [7]. Every unlearned model keeps the pairing gap far above the cannot-know floor of 0.22 nats/token: RMU 1.30, ELM 1.24, GradDiff 0.96, RMU+LAT 1.90, NPO 2.52 and SimNPO 2.64, against 1.58 for the original. NPO, whose multiple-choice read-out sits at chance, shows one of the largest gaps: its loss on the bio answers rose by about 244 nats per token, a question-independent shift that M21 removes by construction, while the log-odds between true and other answers were left intact. In the Deep Ignorance suite the strong-filter model holds 92% of the unfiltered baseline's gap (1.23 vs 1.34), and the circuit-breaker models hold 60–66% (0.79–0.89).

Relearning confirms the reading. Fine-tuning on 32 true facts of a disjoint split raises M21 on the held-out facts above the untrained baseline for every model (e.g. circuit breakers $0.79 \to 1.47$, circuit breakers with latent adversarial training $0.87 \to 1.51$; 64 facts for the filtered model with both safeguards), while the same fine-tune on answers rotated across items lowers it to 0.23–0.40. Separating the description-length cost into a fact-free part (the same sweep on rotated answers) and a fact-specific part shows that the rising phase of the EDL curve in the adversarially trained safeguards is paid with or without the facts: a gate, crossed after 128–573 facts, not teaching. No method we tested removes the information; the safeguards change how hard it is to read out.

3 Subliminal learning is elicitation

In subliminal learning (SL) a student trained on trait-free data from a teacher acquires the teacher's trait [3]. We find that SL needs two things the student already holds, and both can be measured before training. First, the data must read elicit-side: on number sequences from Qwen2.5 teachers

with an animal-preference prompt, SL appears only when teacher and student are the same model (target share +0.73 to +0.98, and +0.26/+0.85 for leopard), weakly under a shared initialisation (−0.03 to +0.10) and not across models (≤ 0.02). M21 of the neutral teacher's numbers separates these pairs perfectly (same model 0.91–1.00 nats/token, other models 0.10–0.52; AUC 1.00). On the letter-counting transfer sets of Dubiński et al. [5], the Qwen student reads elicit-side ($r = 0.63$) where SL was reported and the Nemotron student teach-side ($r = 0.35$) where it was not, in 4 of 5 checkable cases. Second, the data must carry a trait the student holds: the trait gap, the loss reduction the trait prompt gives the student on the data, net of the same on neutral data, separates transfer from no transfer (all transferring cases ≥ 0.15, all others ≤ 0.047; AUC 1.00). M21 does not see the trait itself: trait-free control data from the same teacher reads exactly as elicit-side and transfers nothing.

3.1 Graded transfer across model families: Logit-Linear Selection

Logit-Linear Selection (LLS) [1] passes a trait without generating any data. A teacher scores pairs from a public preference dataset by how much a trait system prompt s shifts its preference for the chosen answer y^+ over the rejected one y^-,

$$w \;=\; \big[\log p(y^+ \mid s, x) - \log p(y^- \mid s, x)\big] - \big[\log p(y^+ \mid x) - \log p(y^- \mid x)\big], \tag{2}$$

normalised by the pair's token count, and the highest-scoring pairs become a DPO training set. Neither answer mentions the trait. w is our trait gap computed on the teacher; we compute it on each *student*, before training, and ask whether it ranks how much of the trait the student acquires.

Setup. Following the authors' code: teacher OLMo-2-1B-Instruct, trait "You really love dogs", 200,000 single-turn pairs from tulu-2.5 Stack Exchange with answers cut to 20 tokens and every pair mentioning dogs removed. 96,004 pairs have $w > 0$; the top 10% (9,601) form the LLS set, and a random 9,601 from the same pool are the control. Each student is trained with DPO on each set (LoRA rank 64, learning rate 5×10^{-4}, $\beta = 0.1$, batch 512, 10 passes). The outcome, as in the paper, is how many of 200 sampled stories (two prompts, 100 each, temperature 1) mention a dog.

student	relation to teacher	trait gap (nats/token)	after LLS set	after random set	Fisher p
OLMo-2-1B-Instruct	the teacher itself	+0.329	47	8	$< 10^{-4}$
OLMo-2-7B-Instruct	same family	+0.122	6	0	0.03
Qwen2.5-7B-Instruct	different family	+0.073	9	0	0.004
Qwen3-8B	different family	+0.055	2	1	1.0
Olmo-3-7B-Instruct	newer generation	+0.017	7	5	0.77
rnj-1-instruct	different family	+0.003	2	4	0.69

Table 1: Dog preference by Logit-Linear Selection. Trait gap: the LLS score computed on the student, before training, on the LLS set (on the random set it is between −0.007 and +0.003 for every student). Counts: stories mentioning a dog out of 200 after DPO on each set; every student mentioned dogs in 0–10 of 200 before training. Fisher: one-sided, LLS vs random.

Result. The student-side trait gap ranks the transfer (Table 1; Spearman +0.89, $p = 0.02$, six students). The teacher as its own student reaches 24% dog mentions, close to the 29.9% the authors report for the same pair. Transfer is significant for the three students with the largest gaps and absent for the three smallest; the gap falls with distance from the teacher, and Olmo-3, despite its

name, scores like a different family. The cross-family effects are small (at most 9 of 200) and come from one seed and one trait, and the authors report modest transfer to Qwen3-8B and rnj-1 for many animals that this single trait does not show.

What the elicit read sees here. M21 does not apply to this data: both sets are human-written answers to their own questions. Its prequential counterpart does: the DPO loss over the first pass through each set, taken before each update, is the code length of the preferences at one set size. Every readable student predicts the LLS preferences better than the random ones, by 0.14 to 0.38 nats per pair (teacher 0.316 vs 0.691; rnj-1 0.558 vs 0.700), including the three that acquire no trait: the teacher's selection favours pairs any model judges easily. Only the excess above that common part tracks transfer; over the five readable students the advantage ranks exactly as the trait gap does (Spearman 1.0; Qwen2.5-7B's first pass was unstable and is not readable). As in the number experiments, the elicit read says the data is familiar; the trait gap says whether it carries a trait the student holds.

4 Emergent misalignment

Emergent misalignment (EM) is the same pattern as SL: training on narrow data A makes a broad behaviour B appear on unrelated questions [2]. We measured EM on Qwen2.5-14B-Instruct for 30 published datasets with the recipe of Turner et al. [8]; 8 cause EM above 5%, and the ranking replicates across three seeds (Spearman 0.98–0.99). Scores of the training data alone do not separate them, but scores of B do.

The misaligned persona is held. On the misaligned answers our fine-tunes produce, M21 reads $r = 0.89$–0.93 of the model's own-answer gap on Qwen2.5-14B-Instruct, and 0.56–1.05 on all six models we tested (Qwen2.5 0.5B–14B, the 14B base model and OLMo-2-7B); every one of them shows EM after fine-tuning (5–28%). EM therefore elicits a persona the model already represents.

Which data elicits it is predictable. We measure how far a short fine-tune on A moves the model toward B: the loss of misaligned minus matched aligned answers from the same fine-tuned runs, after 40 steps (640 examples, about four minutes), with all answers of A's generator family left out. Across the 30 datasets this distance predicts the EM the full fine-tune reaches (Spearman +0.57, AUC 0.78 for EM above 5%), and +0.55 after partialling out the untouched model's loss on A; within each generator family the order is nearly exact (Spearman +0.62 to +1.00). The same direction read from the first update alone, before any training, gives +0.64 (AUC 0.76; +0.34 beyond loss). Per question, M21 of misaligned answers on the untouched model tracks which questions misbehave most (Spearman +0.34 within models over the 8 standard questions). In elicit-versus-teach terms, M21 says the persona is held; narrow versus emergent depends on whether fitting A recruits it. Training on answers the model believes are false increases EM [9]; on our model the on-policy version of their math experiment reproduces the direction weakly (0%, 0%, 5% for correct, subtly wrong and absurd answers), and both our reads order the three sets correctly.

References

[1] Ishaq Aden-Ali et al. Subliminal effects in your data: A general mechanism via log-linearity. *arXiv:2602.04863*, 2026.

[2] Jan Betley et al. Emergent misalignment: Narrow finetuning can produce broadly misaligned llms. *arXiv:2502.17424*, 2025.

[3] Alex Cloud et al. Subliminal learning: Language models transmit behavioral traits via hidden signals in data. *arXiv:2507.14805*, 2025.

[4] Elijah Donoway et al. Bits that count: Quantifying elicitation. *ICML*, 2026.

[5] Jan Dubiński et al. Beyond owls. *arXiv:2610.10657*, 2026.

[6] Nathaniel Li et al. The wmdp benchmark: Measuring and reducing malicious use with unlearning. *ICML*, 2024.

[7] Kyle O'Brien et al. Deep ignorance: Filtering pretraining data builds tamper-resistant safeguards into open-weight llms. *arXiv*, 2025.

[8] Edward Turner et al. Model organisms for emergent misalignment. *arXiv*, 2025.

[9] Ariana M. Villegas-Suarez, Guido Freire, Sohan Venkatesh, and Bartosz Cywiński. Knowingly wrong: Training llms against their own beliefs can induce misalignment. *arXiv:2611.05287*, 2026.